Moses Kolleh Sesay

Sustainability & Climate Analyst

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Open to research collaborations on climate and flood risk, water and ESG

PROFILE

I study what climate risk costs, and what adapting to it is worth.

Geologist by training, sustainability analyst by conviction. I've drilled for water in Sierra Leone, analysed what carries pollution from the world's rivers to the coasts, reviewed disaster-loss data at the UN — and now my PhD is on urban flood risk.

EXPERIENCE

PhD Researcher — Climate-Driven Urban Flood Risk Nov 2026 — Present
Scuola Universitaria Superiore IUSS Pavia

- Assessing how projected extreme rainfall affects urban drainage in Monza and Brianza
- Estimating direct and indirect flood losses, from buildings to business interruption, as expected annual losses
- Weighing adaptation options for the drainage network, for risk-based, climate-resilient design

Researcher — Sustainable Artificial Intelligence Sept 2025 — Feb 2026
Digital Society School, Amsterdam University of Applied Sciences

- In a team, built a framework and prototype weighing generative AI's benefits against its environmental costs
- Helped map AI workloads' Scope 2 and 3 footprint — electricity, hardware, data-centre water — against CSRD/ESRS
- Helped build a decision-support tool for managers setting AI strategy and for the staff who prompt every day
- Anchored in UN SDG 12 (Responsible Consumption) and SDG 13 (Climate Action)

Cohort 7 Member — Sustainability Consulting Accelerator Mar 2025 — Apr 2025
OnePointFive (opf.degree)

- Intensive 8-week programme on Net Zero strategy and decarbonisation
- Frameworks: SBTi, CDP, GHG Protocol, ISO 14000, IFRS S1&S2, TCFD, TNFD
- Hands-on labs: materiality assessments, Scope 1–3 accounting, transition planning

Climate Change Adaptation Intern Jun 2023 — Oct 2023
United Nations Office for Disaster Risk Reduction · Bonn, Germany

- Reviewed disaster-loss data for Caribbean countries feeding the Sendai Framework Monitor
- Documented 54 global hazard information systems for the organisation's catalogue
- Wrote a disaster and climate-risk factsheet for Trinidad and Tobago

Field Operations Manager May 2019 — Sept 2019
Team & Team International · Freetown, Sierra Leone

- Led completion of 14 solar-powered boreholes with storage tanks across Sierra Leone
- Sited boreholes by electrical-resistivity survey: 70% of those drilled struck water, and pumping tests measured their yield
- Streamlined the borehole design process in Excel and QGIS
- Trained local staff on sustainable water-system maintenance

Operations Supervisor Sept 2018 — Apr 2019
Sierra Drilling and Water Supply Enterprise · Freetown, Sierra Leone

- Managed a drilling team of 23
- The team constructed 50 boreholes and rehabilitated 100 hand-dug wells
- Used Excel and QGIS for project management and optimisation

SELECTED PROJECTS

Sustainable AI Framework 2025 — 2026 · Amsterdam, NL
A working decision-support prototype, not a slide deck. The team's prototypes are public on GitHub as Sustainable AI Prototypes, with my role in its README. CHECKABLE FROM OUTSIDE

GAIA — Green AI Assessment 2025 — 2026
The workbook, the web estimator and the reference code give the same answers: 3,984 cross-engine checks pass.
CHECKABLE FROM OUTSIDE

Coastal Water Pollution Dynamics 2023 — 2024 · Wageningen, NL
10,226 sub-basins modelled. A property of the model domain used, not an estimate; the thesis is on request.
NOT CHECKABLE FROM OUTSIDE

Flood-Resilient Wuppertal 2023 · Wuppertal, DE
A complete flood-risk management framework delivered to the municipality. The deliverable is held by the client; nothing here claims it was implemented. NOT CHECKABLE FROM OUTSIDE

UN Disaster Risk Reduction — hazard data Jun 2023 — Oct 2023 · Bonn, DE
54 global hazard information systems identified and documented. A count recorded in the deliverable, which UNDRR holds.
NOT CHECKABLE FROM OUTSIDE

Soft Path Water Management 2020 — 2021 · Changsha, CN
Successfully defended at Hunan University, May 2021. Degree awarded: MSc Industrial Engineering, 2021; the certificate is on request. NOT CHECKABLE FROM OUTSIDE

Groundwater Potential Mapping 2016 — 2017 · Freetown, SL
70% success rate in later drilling at survey-selected sites. From the field records of the boreholes sited with the survey, not a controlled trial. NOT CHECKABLE FROM OUTSIDE

EDUCATION

PhD in Sustainable Development and Climate Change 2026 — 2029
Scuola Universitaria Superiore IUSS Pavia, Italy · National PhD Programme
Human society: socio-economic impacts, policies and institutions · In progress, expected 2029 · Topic: urban flood risk and drainage adaptation

MSc Environmental Sciences 2021 — 2024
Wageningen University & Research, Netherlands
Human-Environment Interaction · Water Systems & Global Change · Thesis: socioeconomic drivers of river export of pollutants worldwide · Wageningen African Scholarship Programme scholar · Average grade 7.8/10 (Dutch scale)

MSc Industrial Engineering 2019 — 2021
Hunan University, Changsha, China
Full Chinese Government (MOFCOM) Scholarship · Thesis: soft path water management — Freetown case study · Project management, operations research, systems analysis

BSc Honours Geology 2013 — 2017
University of Sierra Leone, Freetown
First Class Honours · Dissertation: groundwater potential mapping, geophysical approach · Hydrogeology, geophysics, environmental geology

CERTIFICATES

ESG Specialist Program Sep 2024
Corporate Finance Institute
ESG investing, financial strategy and reporting; risk and opportunity analysis

Data Analytics Training Dec 2023
Masterschool
SQL, Python, Excel, Tableau; real-world business cases

Google Advanced Data Analytics Jan 2024
Google — Coursera
Machine learning and regression models; statistical methods and visualisation

Synergizing DRR & Climate Change Adaptation 2023
UN System Staff College
Disaster risk reduction and adaptation, jointly; Sendai Framework in practice

PUBLIC WORK

- Groundwater Investigation Toolkit (Code, 2026) github.com/moseskolleh/WaterProject
- GAIA Framework 2.2 — Green AI Assessment (Framework, 2026) github.com/moseskolleh/GAIA-Framework-
- climatematch-pipeline — climate extremes for West Africa (Code, 2026) github.com/moseskolleh/climatematch-pipeline
- GloBox A/B test analysis (Analysis, 2023) github.com/moseskolleh/A-B-Testing-at-Globox
- Sustainable AI Prototypes (Code, 2025) github.com/moseskolleh/SustainableAIPrototypes
- EcoPrompt Coach — the maintained tool (promptcoach) (Code, 2026) github.com/moseskolleh/promptcoach
- EcoPrompt Coach — this site's edition, with its evidence ledger (Tool, 2026) on the site: carbon-ai.html
- AI carbon factor set — sources, ranges and review dates (Dataset, 2026) on the site: ai-carbon-data.js
- Field Notes (Essays, 2026) on the site: index.html#notes
- Journey map generator — reproducible cartography (Code, 2026) on the site: scripts/generate-journey-map.js

SKILLS

Python	Pandas · NumPy · Matplotlib — proof: Groundwater toolkit: field data in, reports out, on GitHub
QGIS & ArcGIS	geospatial analysis — proof: Groundwater-potential maps from my BSc dissertation, on request; no public GIS project yet
SQL	PostgreSQL · MySQL — proof: GloBox A/B test: the queries and a statistics notebook, on GitHub
Tableau	visual analytics — proof: GloBox A/B test: its README links a Tableau Public dashboard
Machine Learning	regression · classification — proof: Google Advanced Data Analytics certificate; no public project yet
JavaScript	interactive tools — proof: EcoPrompt Coach: engine, web app and extension, on GitHub
Sustainability	ESG Analysis & Integration · GHG Accounting (Scope 1–3) · Climate Risk Assessment · Water Resource Management · Disaster Risk Reduction
Frameworks & standards	SBTi · CDP · GHG Protocol · TCFD · TNFD · ISO 14000 · IFRS S1&S2 · GRI · CSRD